

SELENIUM FULL MATERIAL

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INTRODUCTION OF AUTOMATION TESTING

Important Java concepts required for selenium:

- Conditions(if, if-else, switch)
- Loops(for, while, do-while, for-each)
- OOps(Inheritance, Polymorphism, Encapsulation, Abstraction)
- Method overloading, overriding
- Constructors
- String
- Type casting, Upcasting
- Code optimization
- Collection(List and Set)

Automation:

- Performing any task by using a tool or machine is called as automation.

Advantages:

1. Save the time.
2. Faster
3. Requires less manual effort
4. Restless.
5. Accuracy will be more
6. Multi-task
7. Requires less human resources

Dis Advantages:

1. Initial investment will be more.
2. It requires constant maintenance
3. It requires additional skill sets.

Automation testing:

- Testing an application by using any automation tools is called as automation testing.

Automation Tool:

- It's a software or an application which is used to automate any applications.
 - Ex: Selenium, QTP, Appium, AutoIT etc,

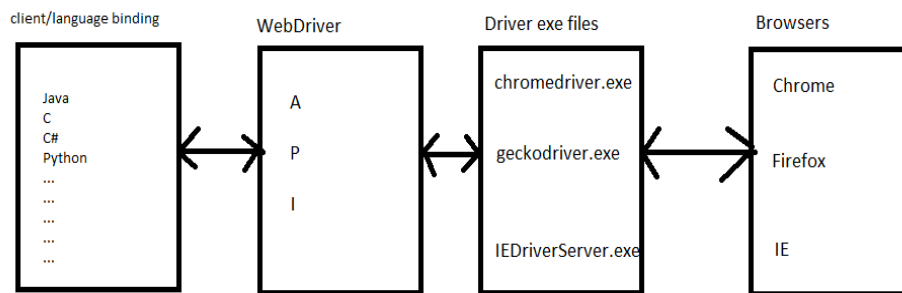
Selenium:

- It's a free and open source automation tool which is used to automation any web based applications.

Advantages of selenium:

- It is freely available automation tool. To make use of selenium for commercial purpose we don't have to buy any license. It is available in below website.
 - <https://www.seleniumhq.org/download/>
- Anyone can view source code of selenium which is available in below website.
 - <https://github.com/SeleniumHQ/selenium>
- Using selenium we can automate any web based applications such as gmail, facebook, flipkart etc...
- It supports for 14 programming languages.
- It supports for multiple platforms such as Windows, Mac, Linux.
- It supports all most all the browsers such as chrome, firefox, ie, safari, opera.

High Level Architecture of selenium:



- Selenium supports for multiple languages such as java, c, python etc,
- Each of these languages contains generic libraries(Re-useable codes) which is called as client/language binding.
- These client binding will communicates with as API called WebDriver
- Here client binding will communicate the actions which has to be performed on the browser such as open the browser, enter url/text, click, close etc.